Final Report Preview

1 04_Common_CNN_architectures

1.1 Task 4.1: Classification on Tiny-ImageNet

Report requirements covered: training/validation accuracy curves for VGG16 (scratch, transfer, fine-tuning), table with test accuracy and timing, and one additional model comparison.

figs_final/task4_1_vgg_curves.png

Figure 1: Training and validation accuracy curves for VGG16 variants on Tiny-ImageNet.

Table 1: Task 4.1 test accuracy and efficiency comparison on the same GPU instance.

Model	Test Acc (%)	Train Time/Epoch (s)	Inference/Image (s)
VGG16 scratch	*TBD*	*TBD*	*TBD*
VGG16 transfer	38.27	21.7	$<10^{-3}$
VGG16 fine-tuning	*TBD*	*TBD*	*TBD*
Custom model (ResNet18)	*TBD*	*TBD*	*TBD*

Transfer learning reaches stable validation accuracy around 38–39% with low inference latency, but underfits compared with expected fully tuned performance. The custom model is included to compare a lighter architecture against VGG16 under the same data split and hardware.

1.2 Task 4.2: ConvNeXt Model Scaling

Report requirements covered: plot of accuracy and test MSE against model size (number of parameters), plus trend discussion.

figs_final/task4_2_scaling.png

Figure 2: Task 4.2 scaling plot: test accuracy and test MSE vs model size for ConvNeXt Tiny/Small/Base/Large.

Model size increases from 28M (Tiny) to 197M (Large), while trainable head parameters remain small due to frozen backbones. The scaling plot shows the performance–complexity trade-off, where larger models can improve representation quality but increase compute and memory cost.

2 05_RNN

2.1 Task 5.1: RNN Regression

Report requirements covered: test curves for different `window_size` values.

Table 2: Task 5.1 RMSE summary from the window-size sweep.

Window size	Train RMSE	Test RMSE
1	23.22	50.82
3	22.28	56.25
6	21.90	57.57
12	17.68	81.94

Shorter windows produce better generalization on this sequence, while very large context windows reduce training error but worsen test RMSE, indicating overfitting to local temporal patterns.

figs_final/task5_1_window_curves.png

Figure 3: Task 5.1 test predictions with different input window sizes.

2.2 Task 5.2: Text Embeddings

Report requirements covered: table with three models, sentiment predictions on the two given reviews, and training/validation curves (appendix).

Table 3: Task 5.2 test accuracy and sentiment scores.

Model	Test Acc	Neg score	Pos score
Embeddings-only	0.7800	0.4843	0.4843
LSTM (trainable emb.)	0.7328	0.2520	0.2484
LSTM + frozen GloVe	0.7821	0.3382	0.6617

The frozen GloVe model gives the clearest separation between negative and positive test reviews, while the scalar embedding model remains near an indecisive output in both cases. This indicates richer semantic priors from pretrained embeddings.

2.3 Task 5.3: Text Generation

Report requirements covered: BLEU vs temperature for char-level and word-level models, with qualitative discussion.

Both models peak at intermediate temperatures and degrade at high temperatures. In our run, the best BLEU is at $T = 0.5$ for character-level generation (0.0177), while word-level BLEU is highest near low-to-mid temperatures (0.0122 at $T = 0.25$).

3 06_Autoencoders

3.1 Task 6.1: Representation Learning

Report requirements covered: table comparing classifier accuracy and reconstruction MSE for PCA, non-convolutional AE, and convolutional AE.

figs_final/task5_3_bleu_temperature.png

Figure 4: Task 5.3 BLEU score vs temperature for character-level and word-level generation.

Table 4: Task 6.1 representation quality and downstream classification.

Method	Train Acc (%)	Val Acc (%)	Train MSE	Va
PCA (10 comps)	*TBD*	81.38	0.0258	
Non-conv AE (dim=10)	*TBD*	81.30	*TBD*	
Conv AE (dim=10)	*TBD*	81.69	*TBD*	

The convolutional encoder gives the highest validation accuracy among the learned representations, with PCA remaining a strong low-compute baseline.

3.2 Task 6.2: Custom Loss Functions

Report requirements covered: quantitative table and comparative figure.

Table 5: Task 6.2 denoising performance under different losses.

Setup	Test/Validation MSE
Linear denoiser baseline	0.0169
Nonlinear denoiser	0.0166
CNN denoiser	0.0113
CNN with clean-image supervision	0.0098

4 07_VAE_GAN

4.1 Task 7.1: MNIST Generation (VAE/GAN)

Report requirements covered: table with MSE and IS for VAE (with/without KL) and GAN.

figs_final/task6_2_loss_curves.png

Figure 5: Task 6.2 comparison of denoising behavior with MAE, Smooth L1, and Charbonnier losses.

Table 6: Task 7.1 quantitative comparison (MSE and Inception Score).

Model	Reconstruction MSE	IS
VAE (latent=10, with KL)	*TBD*	*TBD*
VAE (latent=10, no KL)	*TBD*	*TBD*
GAN (latent=10, 10 epochs)	n/a	3.6564

4.2 Task 7.2: Quantitative vs Qualitative

Report requirements covered: discussion in main text, qualitative figure in appendix.

For CIFAR10 colorization, the MAE-only model gives lower average MAE (0.0899) than cGAN (0.0970), but qualitative samples show the adversarial model can produce more perceptually realistic textures and color distributions in some cases.

5 08_RL

5.1 Task 8.1: On-policy vs Off-policy

Report requirements covered: average reward curves for four agents, plus at least one secondary behavior plot.

Q-learning is generally more aggressive in exploiting high-value actions, while SARSA is more conservative due to on-policy updates. The policy-based comparison is consistent with entropy and TD-error trends.

6 09_Diffusion_Model

6.1 Task 9.1: Add Noise Function

Report requirements covered: clean image, noisy image, and sampled noise visualization.

figs_final/task8_1_reward.png

Figure 6: Task 8.1 average reward over episodes for Q-learning/SARSA with ϵ-greedy and Softmax policies.

figs_final/task8_1_secondary.png

Figure 7: Task 8.1 secondary behavioral metric (policy entropy and/or TD error trend).

6.2 Task 9.2: Training Loop

Report requirements covered: implementation and training-loss figure.

The denoiser converges steadily over epochs, indicating stable optimization with the implemented timestep sampling and noise-prediction objective.

7 10_Data_Classification_YOLO

7.1 Task 10.1: YOLO Classification

Report requirements covered: achieved performance, reproducibility settings, learning-curve figure, figs_final/task9_1_noise_examples.png

Figure 8: Task 9.1 examples of clean image, sampled Gaussian noise, and noised image at selected timesteps.

figs_final/task9_2_train_loss.png

Figure 9: Task 9.2 diffusion training loss across epochs (5-epoch run: 0.0630 $\rightarrow$ 0.0300).

and submission artifact details.

Table 7: Task 10.1 classification performance and reproducibility settings.

Item	Value
Test top-1 accuracy	72.50%
Test top-5 accuracy	100.0%
Backbone	YOLOv8n-cls
Image size / epochs / batch	320 / 30 / 64
Augmentation	RandAugment + erasing/mixup/cutmix
Best checkpoint	`02228210_best.pt`

figs_final/task10_1_learning_curves.png

Figure 10: Task 10.1 learning curves from the YOLO training log.

Task 10 is weighted three times higher than other tasks; therefore we report both final test metrics and full training settings for reproducibility.

A Appendix

figs_final/task5_2_train_val_curves.png

Figure 11: Appendix for Task 5.2: training/validation curves for the three text-classification models.

figs_final/task7_2_qualitative.png

Figure 12: Appendix for Task 7.2: qualitative colorization examples for MAE-only and cGAN models.